

Final Report

Group Members: Ryan Rosiak, Travis Gopaul, Cody Murrer, Keith Flagg, Samuel David

Part 1 General Overview:

Project Description:

This project uses a relational database to store information for Salisbury University's honor societies that will be accessible from each honor society's web page. From faculty advisers, officers and members, to scholarships and more, each society has multiple series of relationships between its entities to consider. This database will organize this data to be accessed, stored, and updated while preserving the data's integrity.

The database will be available to be searched by the general public. Visitors to the site are able to search the database and filter queries for more specific results. With security in mind, sensitive information will only be available to users with login credentials, namely officers, faculty advisers and the super admin. The aforementioned users are able to update, insert, and delete information in the database. The intention is to publish this database online with the ultimate goal being that every honor society at Salisbury University utilizes the database on their respective websites.

The need for this database application stems from the desire to be able to access various points of data pertaining to Salisbury University's honor societies accurately and in a timely manner. As the chapter adviser for Upsilon Pi Epsilon, our client, Dr. Jing, having such a database would be a valuable tool. Ideally, this project will meet this need for chapter advisers of all honor societies at Salisbury University.

Requirements:

- The database keeps track of Salisbury University national **HonorSocieties**, keeping the unique *Name* of the honor society, the *Fees* to be paid when entering the society, the overall *Requirements* to get into the society (i.e. You need a 3.0 GPA or higher to enter), the *Description* of the society, the *OrganizationLink* that leads to the national organization's website, the *Constitution* of the honor society, and the *ByLaws* of the honor society.
- For each of the **HonorSocieties**, there are **Members**. Each member can belong to multiple **HonorSocieties**. The **Members** contain the *Name* of member, the *StudentID* of the member, and the unique *Email* of the member.
- There is a special group of **Members** that are **Officers**. Each officer has a *Title* (i.e. Chapter President), a *YearServed*, and a unique *Email*. Each officer also has an **OfficerLogin** that can be used to manipulate the database.

- Every **HonorSocieties** has (a) **FacultyAdvisors**. Each has a *Name*, a *YearInducted* into the honor society, a unique *Email* associated with the advisor, and a *hsName* (i.e. **HonorSocieties Name**) honor society that they are advisors for. Each advisor also has a **FacultyLogin** that can be used to manipulate the database.
- Each of the **HonorSocieties** awards **Scholarships** to **Members**. Each scholarship has a *Year* given, a unique *Recipient* name, the *Type* of scholarship (i.e. merit based, athletic, cultural, creative, etc), the *Amount* that was given to the *Recipient*, and the *email* (i.e. **Members Email**) of the *Recipient* that received the scholarship.
- Each of the **HonorSocieties** hosts **OrganizationActivities**. Each event has a unique *DateOfActivity*, a *Description* of the activity, the name of the *Speaker* at the activity (i.e. An officer of the chapter or guest speaker), the *ActivityName*, and a unique *hsName* (i.e. **HonorSocieties Name**) that endorses the activity.
- There are **Photos** that are taken of the **OrganizationActivities**. Each photo has a unique *Link* to the picture, the *Year* it was taken, the *DateOfActivity*, and the *hsName* (i.e. **HonorSocieties Name**).
- There are logins for **FacultyAdvisors** and **Officers**. These administrators can create, delete, and update entries in the database. The **FacultyLogin** contains a unique *Username*, *Password*, and an *Email* (i.e. **FacultyAdvisors Email**) associated with the advisor that has that login information. The **OfficerLogin** contains a unique *Username*, *Password*, and an *email* (i.e. **Officers Email**) associated with the officer that has that login information.
- The database contains a linked table **Member-Of** which has a unique *hsName* (i.e. **HonorSocieties Name**), and a unique *email* (i.e. **Members Email**). This table links **Members** to their associated **HonorSocieties**.
- The database contains a linked table **Participates-In** which has a unique *email* (i.e. **Members Email**), a unique *doa* (i.e. **OrganizationActivities DateOfActivity**), and a unique *hsName* (i.e. **HonorSocieties Name**). This table links **Members** to the **OrganizationActivities** that they participate in.

Schema:

HonorSociety(Name VARCHAR(50), Description VARCHAR(500), Requirements VARCHAR(500), Constitution VARCHAR(50), ByLaws VARCHAR(50), OrganizationLink VARCHAR(200), Fees DECIMAL(10,2), primary key(Name));

Members(MembershipID VARCHAR(15), Email VARCHAR(80), Name VARCHAR(50), primary key(Email));

MemberOf(hsName VARCHAR(50), email VARCHAR(80), primary key(hsName, email), foreign key(hsName) references HonorSociety(Name), foreign key(email) references Members(Email));

Scholarship(Recipient VARCHAR(50), Year INT(4), Type VARCHAR(30), Amount DECIMAL(10,2), hsName VARCHAR(50), email VARCHAR(80), primary key(Recipient, Year), foreign key(hsName) references HonorSociety(Name), foreign key(email) references Members(Email));

OrganizationActivities(DateOfActivity DATETIME, Description VARCHAR(200), Speaker VARCHAR(50), ActivityName VARCHAR(50), hsName VARCHAR(50), primary key(DateOfActivity, hsName), foreign key(hsName) references HonorSociety(Name));

ParticipatesIn(email VARCHAR(80), doa DATETIME, hsName VARCHAR(50), primary key(email, doa, hsName), foreign key(email) references Members(Email), foreign key(doa, hsName) references OrganizationActivities(DateOfActivity, hsName));

Photos(Link VARCHAR(50), Year INT(4), doa DATETIME, hsName VARCHAR(50), primary key(Link), foreign key(doa, hsName) references OrganizationActivities(DateOfActivity, hsName));

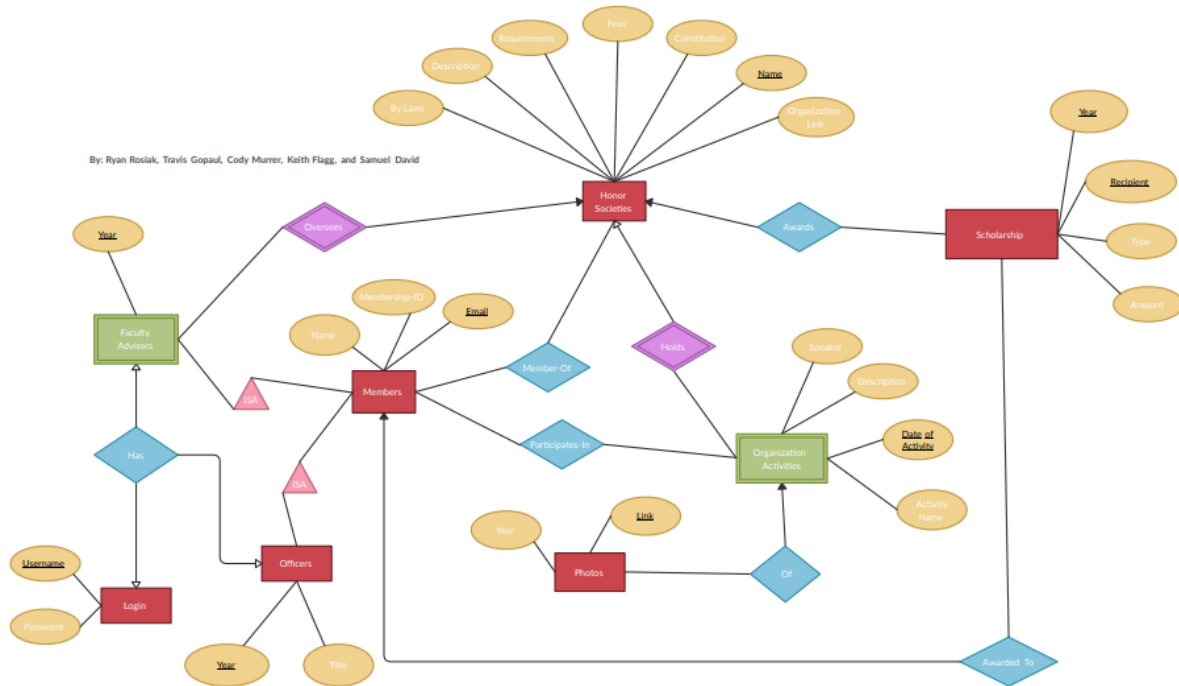
Officers(email VARCHAR(80), Title VARCHAR(50), Year INT(4), primary key(email, Year), foreign key(email) references Members(Email));

FacultyAdvisors(email VARCHAR(80), hsName VARCHAR(50), Year INT(4), primary key(email, hsName, Year), foreign key(hsName) references HonorSociety(Name), foreign key(email) references Members(Email));

FacultyLogin(Username VARCHAR(20), Password VARCHAR(256), email VARCHAR(80), hsName VARCHAR(50), year INT(4), primary key(Username), foreign key(email, hsName, year) references FacultyAdvisors(email, hsName, Year));

OfficerLogin(Username VARCHAR(20), Password VARCHAR(256), email VARCHAR (80), year INT(4), primary key(Username), foreign key(email, year) references Officers(email, Year));

E/R Diagram:



Part 2 Implementation and Testing Details:

Web Link:

<https://lamp.salisbury.edu/~rrosiak1/HonorSocietiesDB/>

Login Credentials:

Username: superadmin

Password: superadminpassword

Implemented Functionality:

- Search Table
- Insert into table (one tuple)
- Delete from table (one tuple)

- Update table (one tuple)
- Login Creation
- Login

Future Functionality:

- Implement session variable for logged in user's honor society
 - This will restrict update/insert/edit capabilities of the user to only be able to make changes to the honor society to which they belong , including if they are in multiple societies
- New user creation notification email
 - Notify the primary faculty advisor whenever a new user is created for the respective honor society
 - Faculty advisor must grant permission for the account to be made
- Forgot/change password functionality
- Allow super admin to be able to change login information of other users
- Ability to add images from the insert page
 - The website currently does not have this capability

Software Packages Used:

- Bootstrap
 - Used to give the application modern styling.
 - Input fields are more modern than in plain CSS
 - Page layout is cleaner as a result of using Bootstrap
- JQuery
 - Used to compare the attributes within dropdowns and hide on-screen attributes.
 - Adds functionality to the UI to create a user friendly experience.

Testing:

On the frontend, we tested the application using developer tools in the browser. On the backend, we tested our application by creating dummy tables and executing queries so that we could affirm that our queries were correct before porting them to the live server. Overall, we all analyzed each other's code for completeness and correctness to ensure that there was no problem left unsolved. All bugs were handled via pair programming or print debugging.

Part 3 Summary:

What We Learned:

- Interface the database with a frontend application
- Inject drop-downs, radio buttons, and checkboxes with PHP
- Gained experience using css for styling
- Created login and new login credentials
- Created a footer and linked different websites such as reddit and twitter.
- Implemented update, insert, and delete interface using PHP
- Gained advanced knowledge MySQL techniques such as cascading
- Had some trouble toughing it out at the end but got together and finished strong
- Many other things!